Bayesian Optimization Report: Function 4

GP Health Summary per Iteration

Iteration	GP Health	Status
1	0.621	Medium
2	0.621	Medium
3	0.620	Medium
4	0.621	Medium
5	0.621	Medium
6	0.622	Medium
7	0.622	Medium
8	0.622	Medium
9	0.623	Medium
10	0.623	Medium
11	0.623	Medium
12	0.624	Medium
13	0.624	Medium
14	0.625	Medium
15	0.625	Medium
16	0.625	Medium
17	0.627	Medium
18	0.628	Medium
19	0.629	Medium
20	0.631	Medium
21	0.632	Medium
22	0.633	Medium
23	0.635	Medium
24	0.636	Medium
25	0.638	Medium
26	0.639	Medium
27	0.640	Medium
28	0.642	Medium
29	0.643	Medium
30	0.645	Medium

GP Sigma Summary

Iteration	Mean Sigma	Min Sigma	Max Sigma
1	2.8915	1.9670	5.0617
2	2.7463	1.9112	5.0753
3	2.6883	1.9129	5.2362
4	2.6137	1.8951	5.0987
5	2.5649	1.8848	4.2309
6	2.5472	1.8107	5.0707
7	2.5227	1.7560	4.7352
8	2.3658	1.6740	4.3778
9	2.4259	1.6741	4.2641
10	2.4599	1.7143	5.0127
11	2.3494	1.6425	4.2603
12	2.2751	1.6035	4.7465
13	2.3235	1.6038	4.0645
14	2.2791	1.5853	4.5082
15	2.2663	1.5405	4.3440
16	1.2610	0.6847	2.3713
17	1.2606	0.6977	2.3445
18	1.2155	0.5190	1.9483
19	1.2137	0.5600	2.1204
20	1.2113	0.5181	2.3970
21	1.1774	0.4461	2.0796
22	1.1724	0.5611	2.0128
23	1.1724	0.5763	2.3291
24	1.1490	0.4305	2.1749
25	1.1666	0.5291	2.1407
26	1.1246	0.3970	2.2336
27	1.1155	0.5020	1.9082
28	1.1308	0.5152	1.9450
29	1.1085	0.5227	1.9955
30	1.1056	0.5587	2.1552

Best Results Per Iteration

Iteration	Best Input	Best Output	Kernel	Acquisition	GP Health	Anomalies
1	[0.52273283 0.42754102 0.02541913 0.10789143]	-12.371612575817558	Matern(ν=0.5)	PORTFOLIO	0.621	None
2	[0.00592553 0.28501306 0.44460667 0.73965787]	-13.323172133708107	Matern(ν=0.5)	PORTFOLIO	0.621	None
3	[0.21905841 0.45774937 0.2225223 0.64621694]	-7.456107146430146	Matern(ν=0.5)	PORTFOLIO	0.620	None
4	[0.13159316 0.45779903 0.59787596 0.35843881]	-7.176594446460935 5	Matern(ν=0.5)	PORTFOLIO	0.621	None
5	[0.56222578 0.23888661 0.39741165 0.61508908]	-6.865009141844292	Matern(ν=0.5)	PORTFOLIO	0.621	None
6	[0.49059771 0.22861558 0.10495153 0.33760546]	-7.837678200960655	Matern(ν=0.5)	PORTFOLIO	0.622	None
7	[0.28738908 0.56084942 0.70876531 0.39213527]	-8.433322970020512	Matern(ν=0.5)	PORTFOLIO	0.622	None
8	[0.546288 0.51559022 0.01684593 0.38708187]	-9.916091157336353	Matern(ν=0.5)	PORTFOLIO	0.622	None
9	[0.28121915 0.25143756 0.22525262 0.12454632]	-8.494699748159114	Matern(ν=0.5)	PORTFOLIO	0.623	None
10	[0.22277563 0.46624339 0.07185979 0.48704282]	-8.276620390317843	Matern(ν=0.5)	PORTFOLIO	0.623	None
11	[0.4150349 0.11374392 0.56248517 0.41247048]	-7.812569412073957	Matern(ν=0.5)	PORTFOLIO	0.623	None
12	[0.4796996 0.6730786 0.51728852 0.27066703]	-7.060595944328737	Matern(ν=0.5)	PORTFOLIO	0.624	None
13	[0.43766812 0.37212902 0.71527794 0.18054017]	-9.459937136511856	Matern(ν=0.5)	PORTFOLIO	0.624	None
14	[0.23907784 0.33113323 0.6023621 0.69511636]	-8.631596402757367	Matern(ν=0.5)	PORTFOLIO	0.625	None
15	[0.36679732 0.72623999 0.22165572 0.45035475]	-8.056165707444926	Matern(ν=0.5)	PORTFOLIO	0.625	None
16	[0.31077885 0.46434819 0.37394311 0.38565443]	-0.663761474789863 5	Matern(ν=0.5)	PORTFOLIO	0.625	None
17	[0.36460513 0.45853382 0.38723057 0.45795651]	-0.155098607881082 98	Matern(ν=0.5)	PORTFOLIO	0.627	None
18	[0.44301794 0.38812514 0.33236708 0.42359818]	-0.502659180031845	Matern(ν=0.5)	PORTFOLIO	0.628	None
19	[0.4027741 0.47152617 0.38129334 0.36631498]	-0.008084815238188 27	Matern(ν=0.5)	PORTFOLIO	0.629	None
20	[0.46175747 0.46039928 0.39938168 0.35999151]	-0.324022398216797 57	Matern(ν=0.5)	PORTFOLIO	0.631	None
21	[0.42291705 0.48596609 0.37893055 0.41775188]	0.020537947381040 31	Matern(ν=0.5)	PORTFOLIO	0.632	None

22	[0.37223216 0.40711247 0.39577289 0.39557243]	0.18038388293525465	Matern(ν=0.5)	PORTFOLIO	0.633	None
23	[0.36126797 0.44256197 0.44774761 0.34710121]	-0.4542843235685883	Matern(ν=0.5)	PORTFOLIO	0.635	None
24	[0.36705821 0.39232699 0.41858157 0.43172931]	-0.2456548719791698	Matern(ν=0.5)	PORTFOLIO	0.636	None
25	[0.29391695 0.46004579 0.41904184 0.36854533]	-0.9678974576281281	Matern(ν=0.5)	PORTFOLIO	0.638	None
26	[0.41726458 0.43625573 0.4849769 0.3380734]	-0.8863099572876472	Matern(ν=0.5)	PORTFOLIO	0.639	None
27	[0.43096116 0.43321671 0.40880145 0.41702186]	0.37778791185071725	Matern(ν=0.5)	PORTFOLIO	0.640	None
28	[0.38612598 0.3809956 0.40302495 0.4003041]	-0.06022327928271487	Matern(ν=0.5)	PORTFOLIO	0.642	None
29	[0.45778664 0.44625198 0.40458958 0.39450648]	0.049536669818708745	Matern(ν=0.5)	PORTFOLIO	0.643	None
30	[0.40302758 0.510745 0.46079316 0.41744523]	-0.5607760649286213	Matern(ν=0.5)	PORTFOLIO	0.645	None

Acquisition Values Per Iteration

Iteration 1

Method	Max Value
UCB	0.262405
EI	0.000305
PI	0.000472
THOMPSON	-1.937536
KG	0.000305
ENTROPY	339.206404
PORTFOLIO	4.330672

Iteration 2

Method	Max Value
UCB	-2.235246
EI	0.000294
PI	0.000501
THOMPSON	-2.797068
KG	0.000294
ENTROPY	258.304357

PORTFOLIO	3.485483

Iteration 3

Method	Max Value
UCB	-2.924002
EI	0.000025
PI	0.000048
THOMPSON	-1.746106
KG	0.000025
ENTROPY	276.323203
PORTFOLIO	1.733957

Iteration 4

Method	Max Value
UCB	-2.902267
EI	0.000021
PI	0.000043
THOMPSON	-2.618280
KG	0.000021
ENTROPY	318.253379
PORTFOLIO	0.764161

Iteration 5

Method	Max Value
UCB	-2.836569
EI	0.000018
PI	0.000038
THOMPSON	-3.375653
KG	0.000018
ENTROPY	430.355685
PORTFOLIO	0.889917

Iteration 6

Method	Max Value
UCB	-3.124462

EI	0.000004
PI	0.000011
THOMPSON	-3.593076
KG	0.000004
ENTROPY	378.104073
PORTFOLIO	0.509404

Iteration 7

Method	Max Value
UCB	-3.914436
EI	0.000003
PI	0.000005
THOMPSON	-4.439938
KG	0.000003
ENTROPY	339.763785
PORTFOLIO	0.227041

Iteration 8

Method	Max Value
UCB	-1.985674
EI	0.000094
PI	0.000212
THOMPSON	-2.134266
KG	0.000094
ENTROPY	270.655904
PORTFOLIO	0.346152

Iteration 9

Method	Max Value
UCB	-2.579281
EI	0.000008
PI	0.000020
THOMPSON	-4.226103
KG	0.000008
ENTROPY	326.133692

PORTFOLIO	-0.051484

Iteration 10

Method	Max Value
UCB	-3.477111
EI	0.000001
PI	0.000002
THOMPSON	-4.118252
KG	0.000001
ENTROPY	302.204215
PORTFOLIO	-0.726607

Iteration 11

Method	Max Value
UCB	-3.620393
EI	0.000000
PI	0.000000
THOMPSON	-4.758638
KG	0.000000
ENTROPY	430.837367
PORTFOLIO	-0.992479

Iteration 12

Method	Max Value
UCB	-3.200393
EI	0.000000
PI	0.000001
THOMPSON	-2.850083
KG	0.000000
ENTROPY	267.654503
PORTFOLIO	-0.831735

Iteration 13

Method	Max Value
UCB	-4.673998

EI	0.000000
PI	0.000000
THOMPSON	-4.271093
KG	0.000000
ENTROPY	387.512501
PORTFOLIO	-1.577973

Iteration 14

Method	Max Value
UCB	-4.074209
EI	0.000000
PI	0.000000
THOMPSON	-4.617838
KG	0.000000
ENTROPY	inf
PORTFOLIO	-1.730532

Iteration 15

Method	Max Value
UCB	-3.820750
EI	0.000000
PI	0.000000
THOMPSON	-5.283655
KG	0.000000
ENTROPY	387.907038
PORTFOLIO	-1.586553

Iteration 16

Method	Max Value
UCB	0.673936
EI	0.085192
PI	0.162583
THOMPSON	1.765272
KG	0.085192
ENTROPY	22.205759

PORTFOLIO	1.031034

Iteration 17

Method	Max Value
UCB	0.531632
EI	0.052880
PI	0.116046
THOMPSON	1.294450
KG	0.052880
ENTROPY	34.220022
PORTFOLIO	0.817426

Iteration 18

Method	Max Value
UCB	0.628950
EI	0.072750
PI	0.154572
THOMPSON	0.616983
KG	0.072750
ENTROPY	81.358491
PORTFOLIO	1.055054

Iteration 19

Method	Max Value
UCB	0.627652
EI	0.072838
PI	0.148750
THOMPSON	0.817143
KG	0.072838
ENTROPY	50.167875
PORTFOLIO	0.971257

Iteration 20

Method	Max Value
UCB	0.648071

EI	0.079621
PI	0.226130
THOMPSON	0.650242
KG	0.079621
ENTROPY	73.786859
PORTFOLIO	0.923106

Iteration 21

Method	Max Value
UCB	0.612278
EI	0.069106
PI	0.227538
THOMPSON	0.416039
KG	0.069106
ENTROPY	45.886858
PORTFOLIO	0.648895

Iteration 22

Method	Max Value
UCB	0.624121
EI	0.069719
PI	0.178048
THOMPSON	0.614291
KG	0.069719
ENTROPY	51.813752
PORTFOLIO	0.605470

Iteration 23

Method	Max Value
UCB	0.557737
EI	0.053028
PI	0.141194
THOMPSON	0.606881
KG	0.053028
ENTROPY	42.061046

PORTFOLIO	0.627737

Iteration 24

Method	Max Value
UCB	0.427046
EI	0.028104
PI	0.090058
THOMPSON	0.821817
KG	0.028104
ENTROPY	49.218212
PORTFOLIO	0.643765

Iteration 25

Method	Max Value
UCB	0.488607
EI	0.037465
PI	0.115462
THOMPSON	0.458435
KG	0.037465
ENTROPY	43.022192
PORTFOLIO	0.688813

Iteration 26

Method	Max Value
UCB	0.407216
EI	0.019401
PI	0.088283
THOMPSON	0.244840
KG	0.019401
ENTROPY	40.995838
PORTFOLIO	0.408418

Iteration 27

Method	Max Value
UCB	0.628535

EI	0.071191
PI	0.214412
THOMPSON	0.629640
KG	0.071191
ENTROPY	45.318982
PORTFOLIO	0.531836

Iteration 28

Method	Max Value
UCB	0.492718
EI	0.036863
PI	0.119853
THOMPSON	0.497918
KG	0.036863
ENTROPY	46.512689
PORTFOLIO	0.436983

Iteration 29

Method	Max Value
UCB	0.503121
EI	0.037766
PI	0.128812
THOMPSON	0.345688
KG	0.037766
ENTROPY	50.627411
PORTFOLIO	0.473180

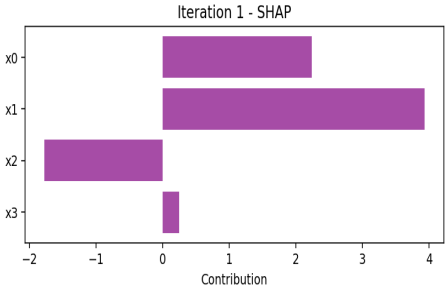
Iteration 30

Method	Max Value
UCB	0.417608
EI	0.024560
PI	0.087008
THOMPSON	0.646104
KG	0.024560
ENTROPY	80.777384

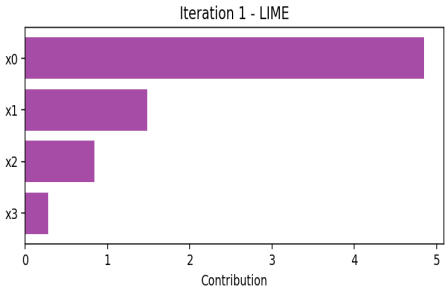
PORTFOLIO	0.473198

Feature Explanations (SHAP / LIME)

Iteration 1 - SHAP



Iteration 1 - LIME



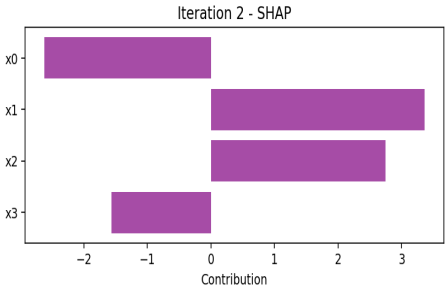
Iteration 1 - SHAP Values

Feature	SHAP Value
x0	2.23478
x1	3.93572
x2	-1.77820
x3	0.24951

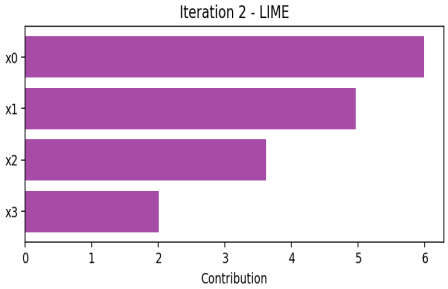
Iteration 1 - LIME Values

Feature	LIME Value
x0	4.84510
x1	1.48099
x2	0.83965
x3	0.27518

Iteration 2 - SHAP



Iteration 2 - LIME



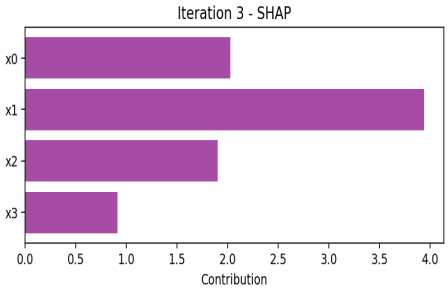
Iteration 2 - SHAP Values

Feature	SHAP Value
x0	-2.62161
x1	3.36408
x2	2.74674
x3	-1.56785

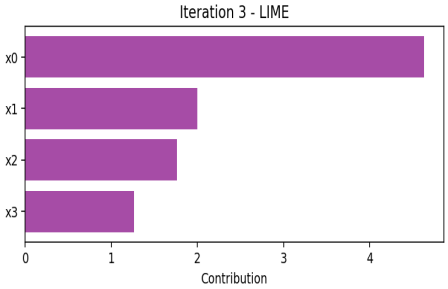
Iteration 2 - LIME Values

Feature	LIME Value
x0	5.98647
x1	4.96108
x2	3.61380
x3	2.00103

Iteration 3 - SHAP



Iteration 3 - LIME



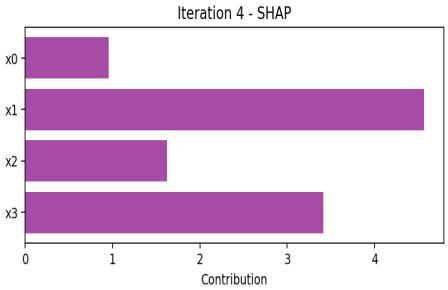
Iteration 3 - SHAP Values

Feature	SHAP Value
x0	2.02709
x1	3.93882
x2	1.90054
x3	0.91372

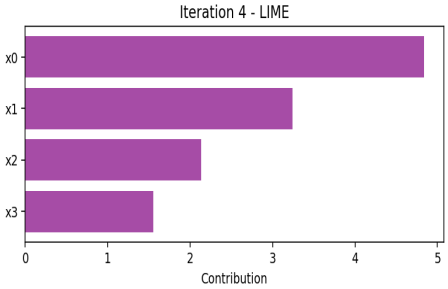
Iteration 3 - LIME Values

Feature	LIME Value
x0	4.63164
x1	1.99756
x2	1.75921
x3	1.26585

Iteration 4 - SHAP



Iteration 4 - LIME



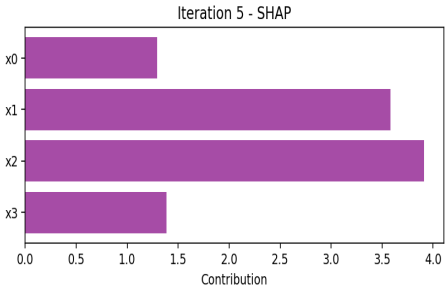
Iteration 4 - SHAP Values

Feature	SHAP Value
x0	0.95262
x1	4.56455
x2	1.62168
x3	3.41158

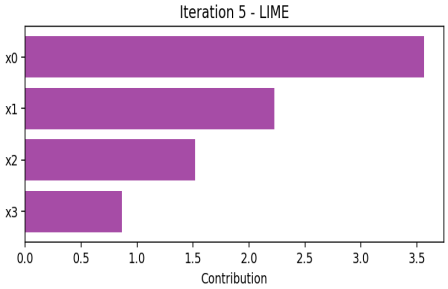
Iteration 4 - LIME Values

Feature	LIME Value
x0	4.83468
x1	3.24172
x2	2.13025
x3	1.55293

Iteration 5 - SHAP



Iteration 5 - LIME



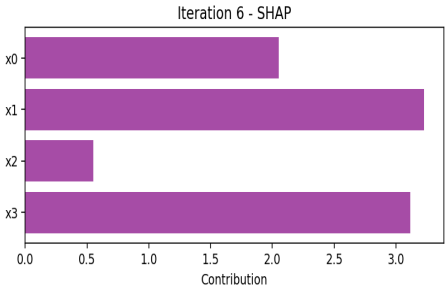
Iteration 5 - SHAP Values

Feature	SHAP Value
x0	1.29044
x1	3.58150
x2	3.90951
x3	1.38213

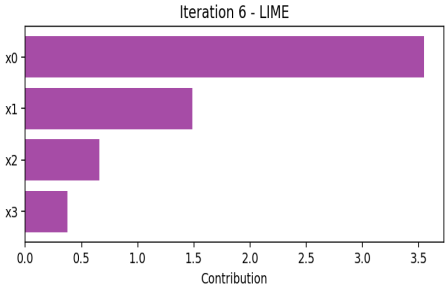
Iteration 5 - LIME Values

Feature	LIME Value
x0	3.56385
x1	2.22854
x2	1.51887
x3	0.86497

Iteration 6 - SHAP



Iteration 6 - LIME



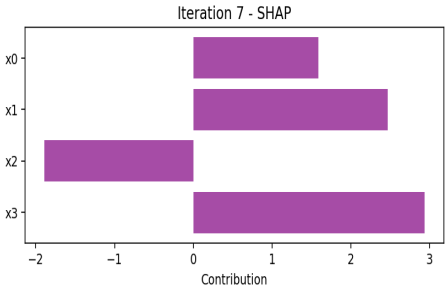
Iteration 6 - SHAP Values

Feature	SHAP Value
x0	2.05207
x1	3.22585
x2	0.55088
x3	3.11634

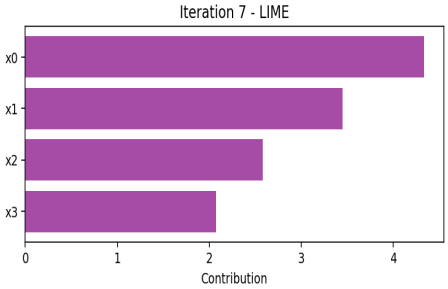
Iteration 6 - LIME Values

Feature	LIME Value
x0	3.54530
x1	1.48508
x2	0.65885
x3	0.37626

Iteration 7 - SHAP



Iteration 7 - LIME



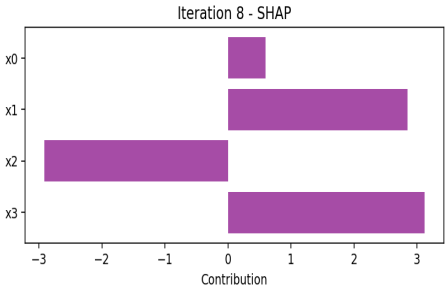
Iteration 7 - SHAP Values

Feature	SHAP Value
x0	1.59044
x1	2.47095
x2	-1.89114
x3	2.93953

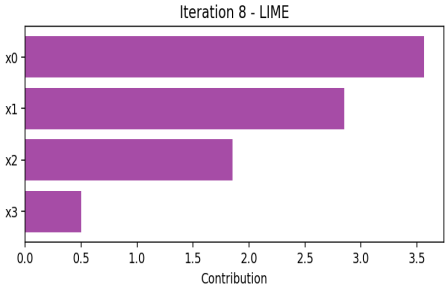
Iteration 7 - LIME Values

Feature	LIME Value
x0	4.33091
x1	3.44821
x2	2.58250
x3	2.07441

Iteration 8 - SHAP



Iteration 8 - LIME



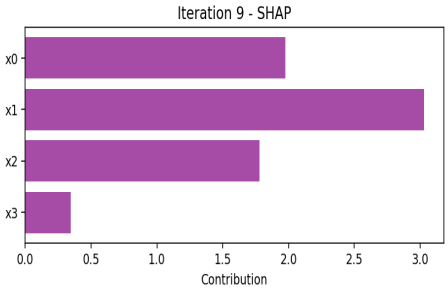
Iteration 8 - SHAP Values

Feature	SHAP Value
x0	0.59977
x1	2.84917
x2	-2.91410
x3	3.12523

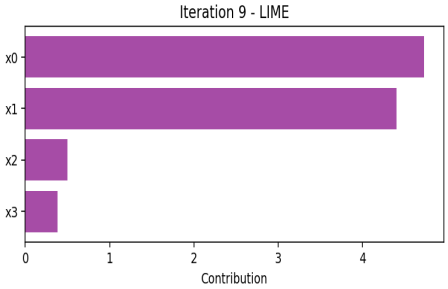
Iteration 8 - LIME Values

Feature	LIME Value
x0	3.56250
x1	2.85001
x2	1.85258
x3	0.49698

Iteration 9 - SHAP



Iteration 9 - LIME



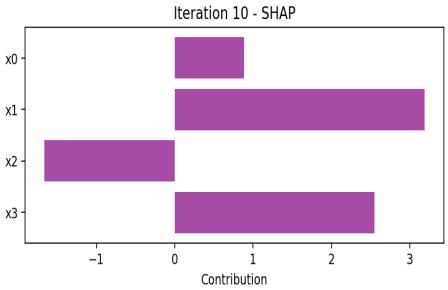
Iteration 9 - SHAP Values

Feature	SHAP Value
x0	1.97764
x1	3.02899
x2	1.77779
x3	0.34556

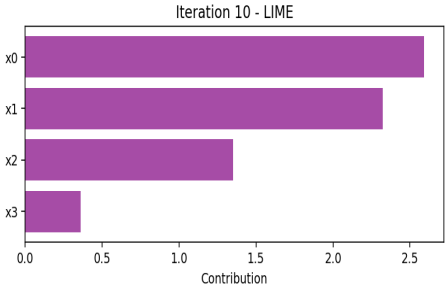
Iteration 9 - LIME Values

Feature	LIME Value
x0	4.72806
x1	4.40229
x2	0.50008
x3	0.37899

Iteration 10 - SHAP



Iteration 10 - LIME



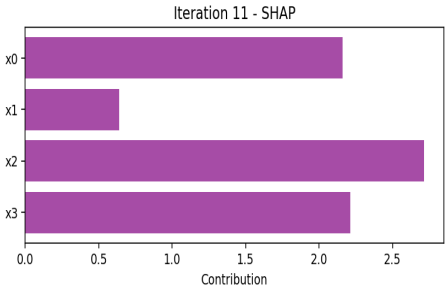
Iteration 10 - SHAP Values

Feature	SHAP Value
x0	0.88653
x1	3.19346
x2	-1.66592
x3	2.55027

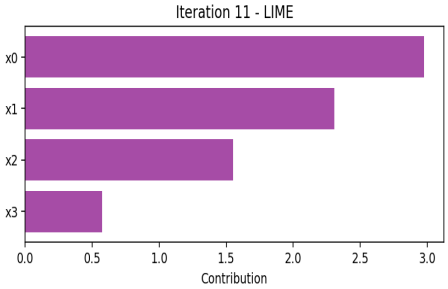
Iteration 10 - LIME Values

Feature	LIME Value
x0	2.59053
x1	2.32484
x2	1.34943
x3	0.35883

Iteration 11 - SHAP



Iteration 11 - LIME



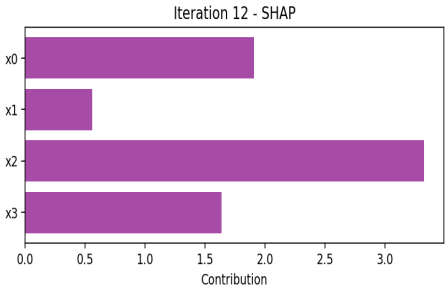
Iteration 11 - SHAP Values

Feature	SHAP Value
x0	2.15926
x1	0.63986
x2	2.71361
x3	2.21098

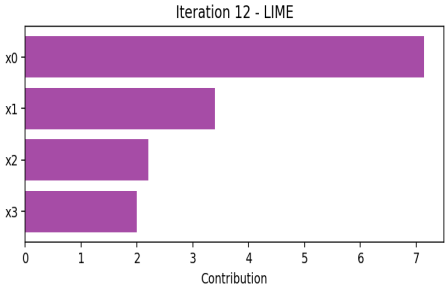
Iteration 11 - LIME Values

Feature	LIME Value
x0	2.97600
x1	2.30663
x2	1.55080
x3	0.57403

Iteration 12 - SHAP



Iteration 12 - LIME



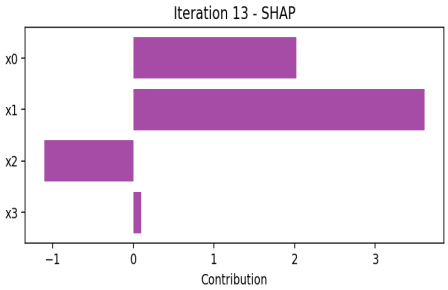
Iteration 12 - SHAP Values

Feature	SHAP Value
x0	1.90682
x1	0.55693
x2	3.32434
x3	1.63844

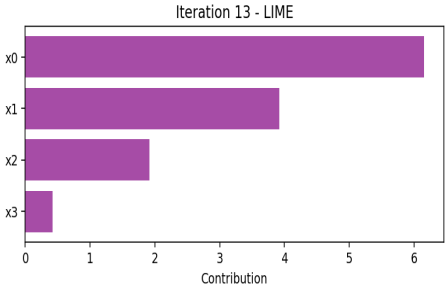
Iteration 12 - LIME Values

Feature	LIME Value
x0	7.13698
x1	3.39640
x2	2.20155
x3	1.99192

Iteration 13 - SHAP



Iteration 13 - LIME



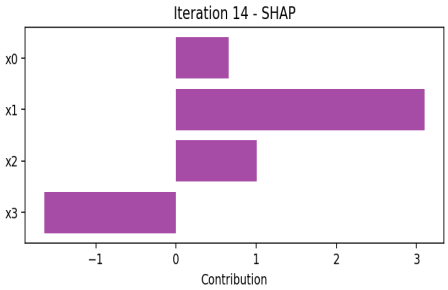
Iteration 13 - SHAP Values

Feature	SHAP Value
x0	2.02044
x1	3.61486
x2	-1.10442
x3	0.09649

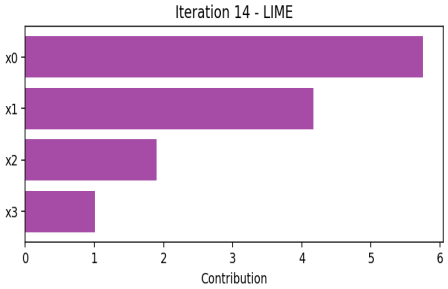
Iteration 13 - LIME Values

Feature	LIME Value
x0	6.15438
x1	3.91936
x2	1.91408
x3	0.42339

Iteration 14 - SHAP



Iteration 14 - LIME



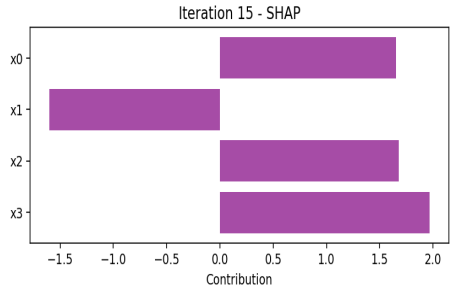
Iteration 14 - SHAP Values

Feature	SHAP Value
x0	0.65658
x1	3.10366
x2	1.00878
x3	-1.64287

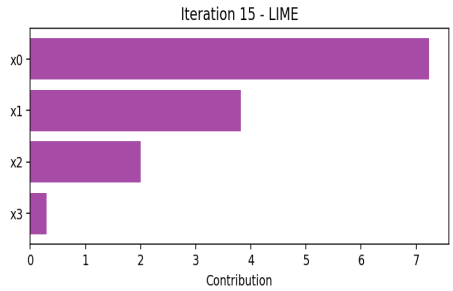
Iteration 14 - LIME Values

Feature	LIME Value
x0	5.75707
x1	4.16600
x2	1.89764
x3	1.00915

Iteration 15 - SHAP



Iteration 15 - LIME



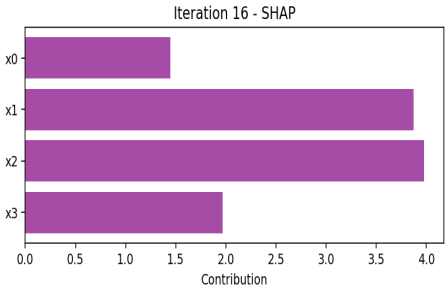
Iteration 15 - SHAP Values

Feature	SHAP Value
x0	1.65182
x1	-1.60092
x2	1.68092
x3	1.96977

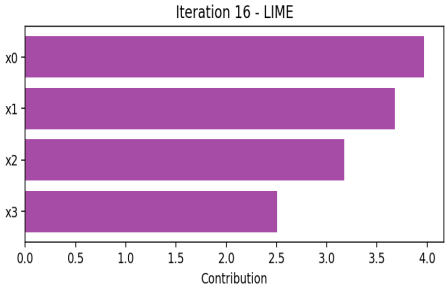
Iteration 15 - LIME Values

Feature	LIME Value
x0	7.22987
x1	3.82034
x2	2.00411
x3	0.29862

Iteration 16 - SHAP



Iteration 16 - LIME



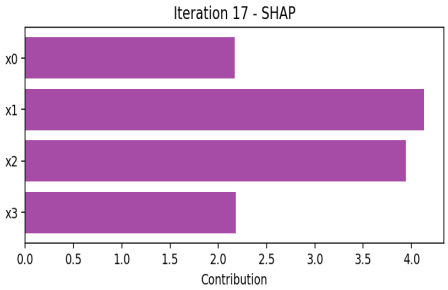
Iteration 16 - SHAP Values

Feature	SHAP Value
x0	1.44883
x1	3.86947
x2	3.97454
x3	1.96504

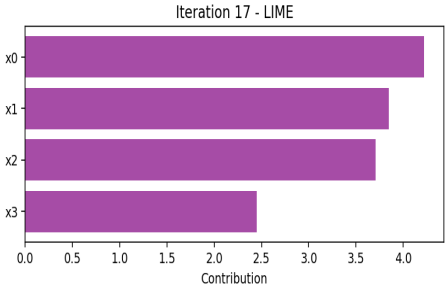
Iteration 16 - LIME Values

Feature	LIME Value
x0	3.96725
x1	3.68009
x2	3.17342
x3	2.50731

Iteration 17 - SHAP



Iteration 17 - LIME



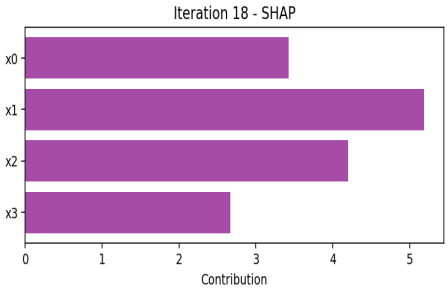
Iteration 17 - SHAP Values

Feature	SHAP Value
x0	2.16758
x1	4.12934
x2	3.94317
x3	2.17984

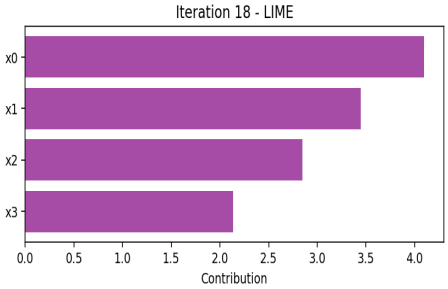
Iteration 17 - LIME Values

Feature	LIME Value
x0	4.21894
x1	3.84508
x2	3.70654
x3	2.45083

Iteration 18 - SHAP



Iteration 18 - LIME



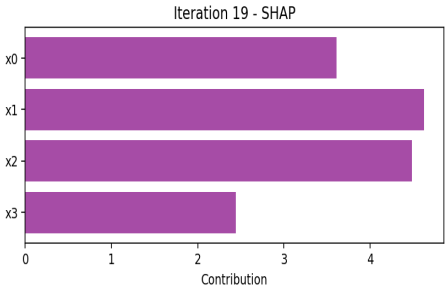
Iteration 18 - SHAP Values

Feature	SHAP Value
x0	3.42373
x1	5.18543
x2	4.19659
x3	2.66935

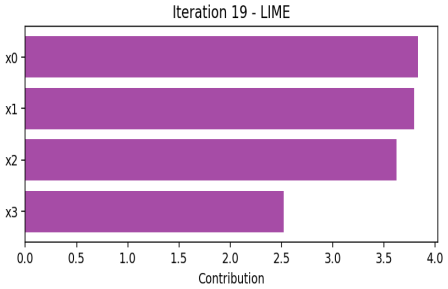
Iteration 18 - LIME Values

Feature	LIME Value
x0	4.09503
x1	3.44595
x2	2.84888
x3	2.13231

Iteration 19 - SHAP



Iteration 19 - LIME



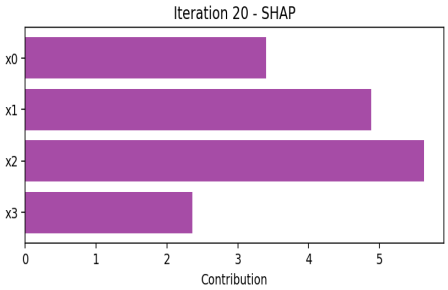
Iteration 19 - SHAP Values

Feature	SHAP Value
x0	3.61275
x1	4.62428
x2	4.48871
x3	2.44507

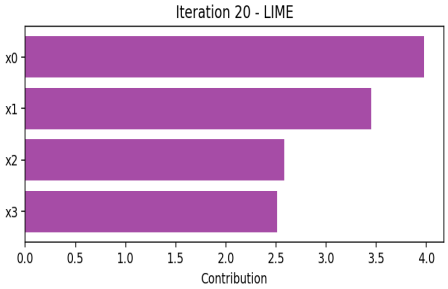
Iteration 19 - LIME Values

Feature	LIME Value
x0	3.83264
x1	3.79819
x2	3.62609
x3	2.52002

Iteration 20 - SHAP



Iteration 20 - LIME



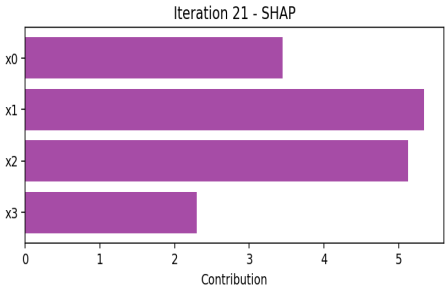
Iteration 20 - SHAP Values

Feature	SHAP Value
x0	3.39915
x1	4.88421
x2	5.62493
x3	2.35817

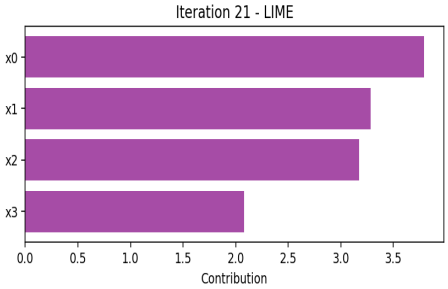
Iteration 20 - LIME Values

Feature	LIME Value
x0	3.97372
x1	3.45127
x2	2.58230
x3	2.50997

Iteration 21 - SHAP



Iteration 21 - LIME



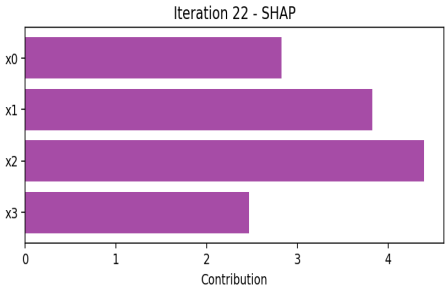
Iteration 21 - SHAP Values

Feature	SHAP Value
x0	3.44398
x1	5.33811
x2	5.12518
x3	2.29297

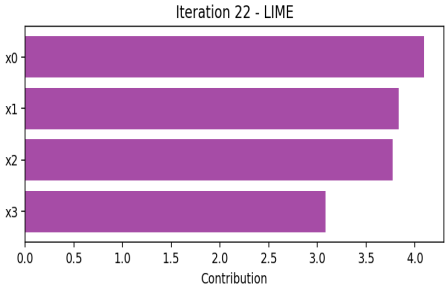
Iteration 21 - LIME Values

Feature	LIME Value
x0	3.78934
x1	3.28564
x2	3.17604
x3	2.07996

Iteration 22 - SHAP



Iteration 22 - LIME



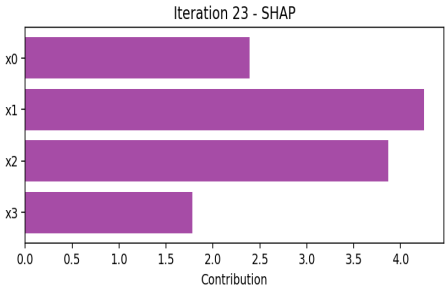
Iteration 22 - SHAP Values

Feature	SHAP Value
x0	2.82422
x1	3.82725
x2	4.39540
x3	2.46882

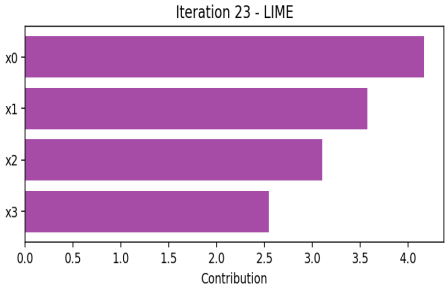
Iteration 22 - LIME Values

Feature	LIME Value
x0	4.09340
x1	3.83774
x2	3.77035
x3	3.08453

Iteration 23 - SHAP



Iteration 23 - LIME



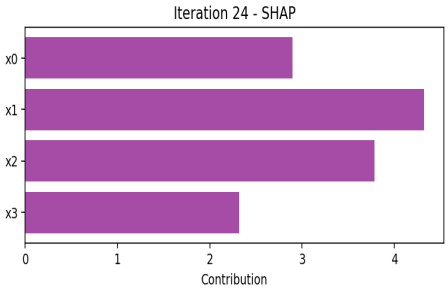
Iteration 23 - SHAP Values

Feature	SHAP Value
x0	2.39066
x1	4.24754
x2	3.87087
x3	1.77948

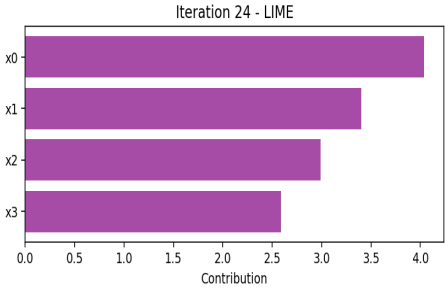
Iteration 23 - LIME Values

Feature	LIME Value
x0	4.16544
x1	3.57232
x2	3.10198
x3	2.54618

Iteration 24 - SHAP



Iteration 24 - LIME



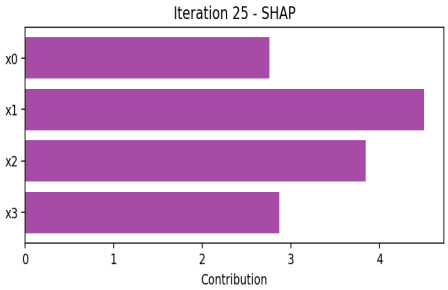
Iteration 24 - SHAP Values

Feature	SHAP Value
x0	2.89670
x1	4.32157
x2	3.78347
x3	2.32039

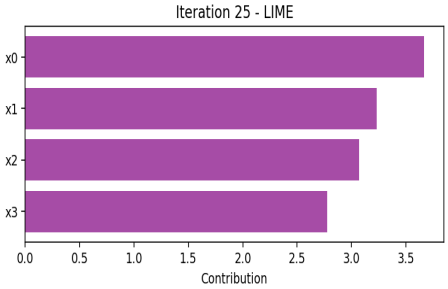
Iteration 24 - LIME Values

Feature	LIME Value
x0	4.03532
x1	3.40131
x2	2.99226
x3	2.58559

Iteration 25 - SHAP



Iteration 25 - LIME



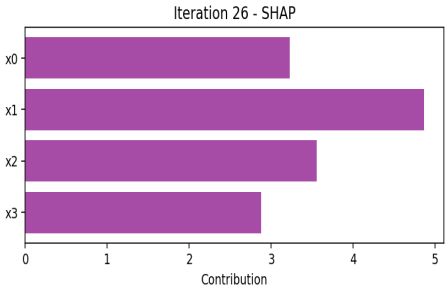
Iteration 25 - SHAP Values

Feature	SHAP Value
x0	2.75979
x1	4.50150
x2	3.84391
x3	2.86964

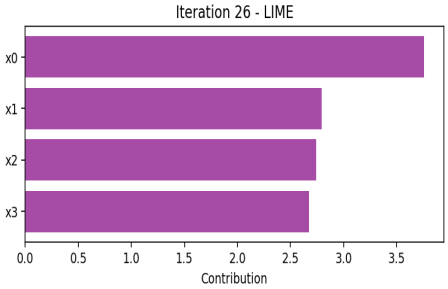
Iteration 25 - LIME Values

Feature	LIME Value
x0	3.66571
x1	3.23255
x2	3.07059
x3	2.77410

Iteration 26 - SHAP



Iteration 26 - LIME



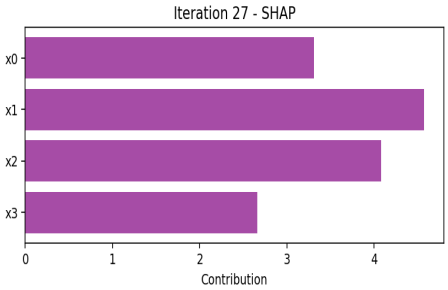
Iteration 26 - SHAP Values

Feature	SHAP Value
x0	3.22355
x1	4.86288
x2	3.55563
x3	2.87593

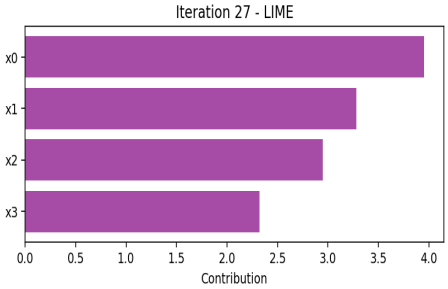
Iteration 26 - LIME Values

Feature	LIME Value
x0	3.75758
x1	2.79477
x2	2.74195
x3	2.67627

Iteration 27 - SHAP



Iteration 27 - LIME



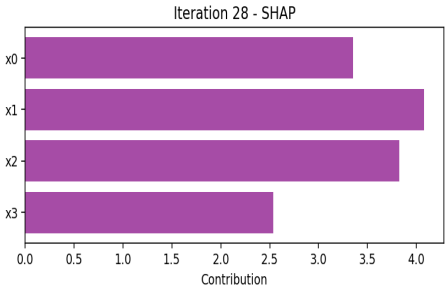
Iteration 27 - SHAP Values

Feature	SHAP Value
x0	3.30877
x1	4.57034
x2	4.08046
x3	2.65827

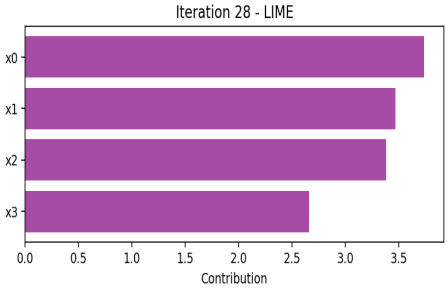
Iteration 27 - LIME Values

Feature	LIME Value
x0	3.95102
x1	3.28301
x2	2.94901
x3	2.32238

Iteration 28 - SHAP



Iteration 28 - LIME



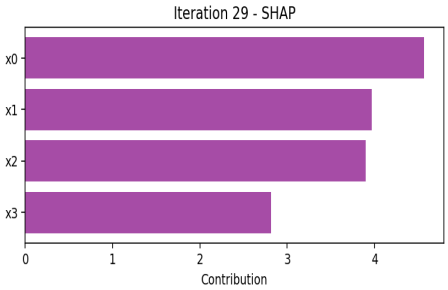
Iteration 28 - SHAP Values

Feature	SHAP Value
x0	3.35300
x1	4.07733
x2	3.82775
x3	2.53593

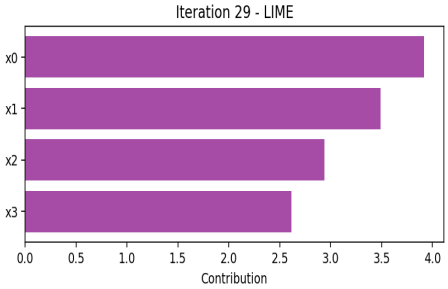
Iteration 28 - LIME Values

Feature	LIME Value
x0	3.73520
x1	3.47043
x2	3.37902
x3	2.65985

Iteration 29 - SHAP



Iteration 29 - LIME



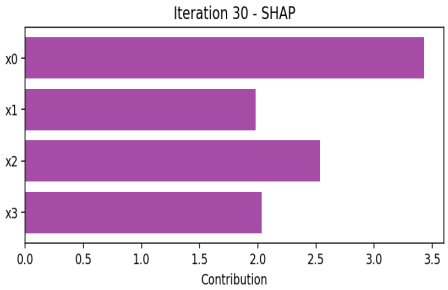
Iteration 29 - SHAP Values

Feature	SHAP Value
x0	4.56547
x1	3.96785
x2	3.89896
x3	2.81583

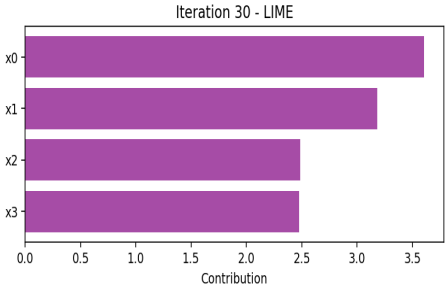
Iteration 29 - LIME Values

Feature	LIME Value
x0	3.91627
x1	3.49103
x2	2.94080
x3	2.61223

Iteration 30 - SHAP



Iteration 30 - LIME



Iteration 30 - SHAP Values

Feature	SHAP Value
x0	3.42798
x1	1.97974
x2	2.53550
x3	2.03269

Iteration 30 - LIME Values

Feature	LIME Value
x0	3.60367
x1	3.18368
x2	2.48652
x3	2.47426